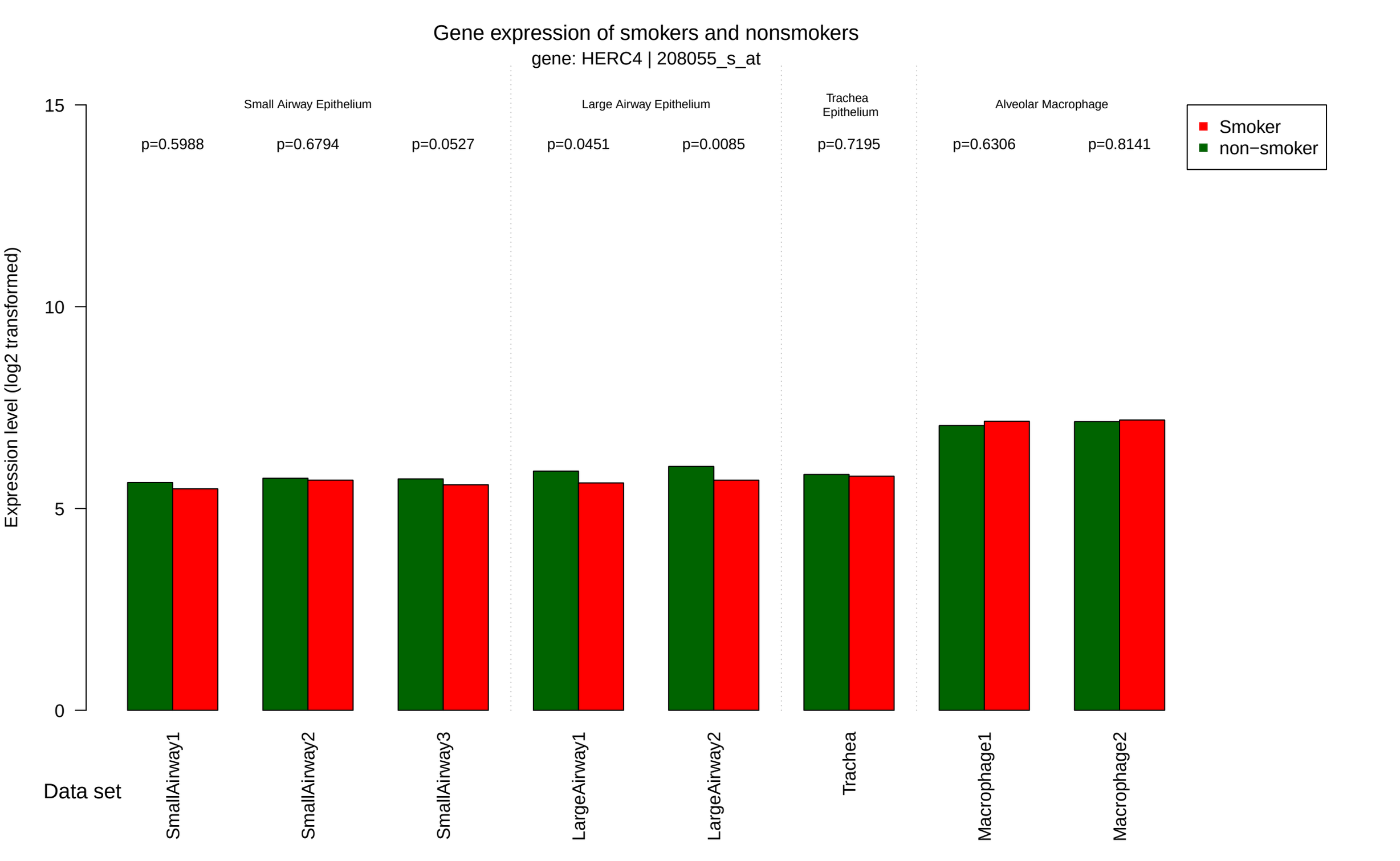
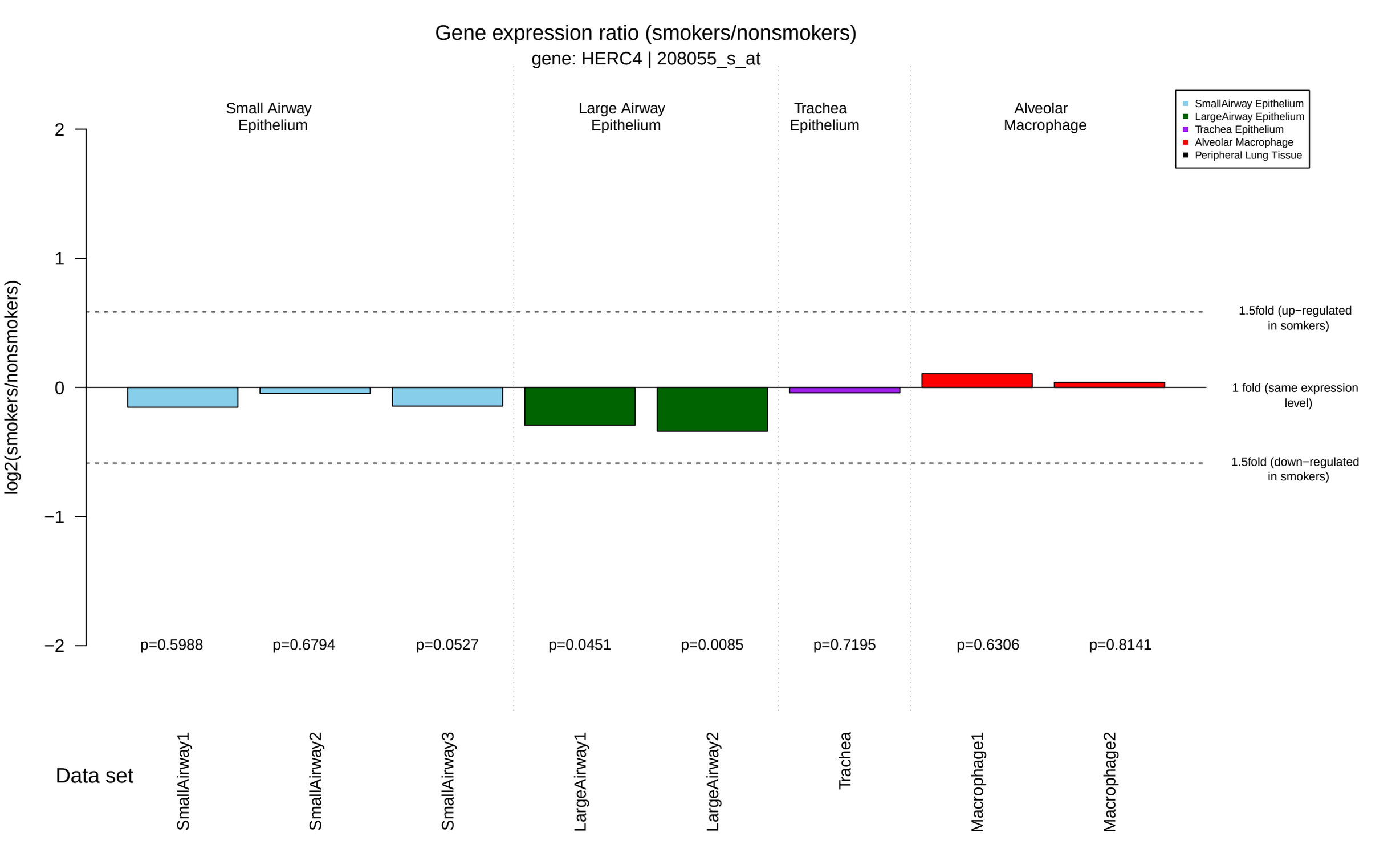
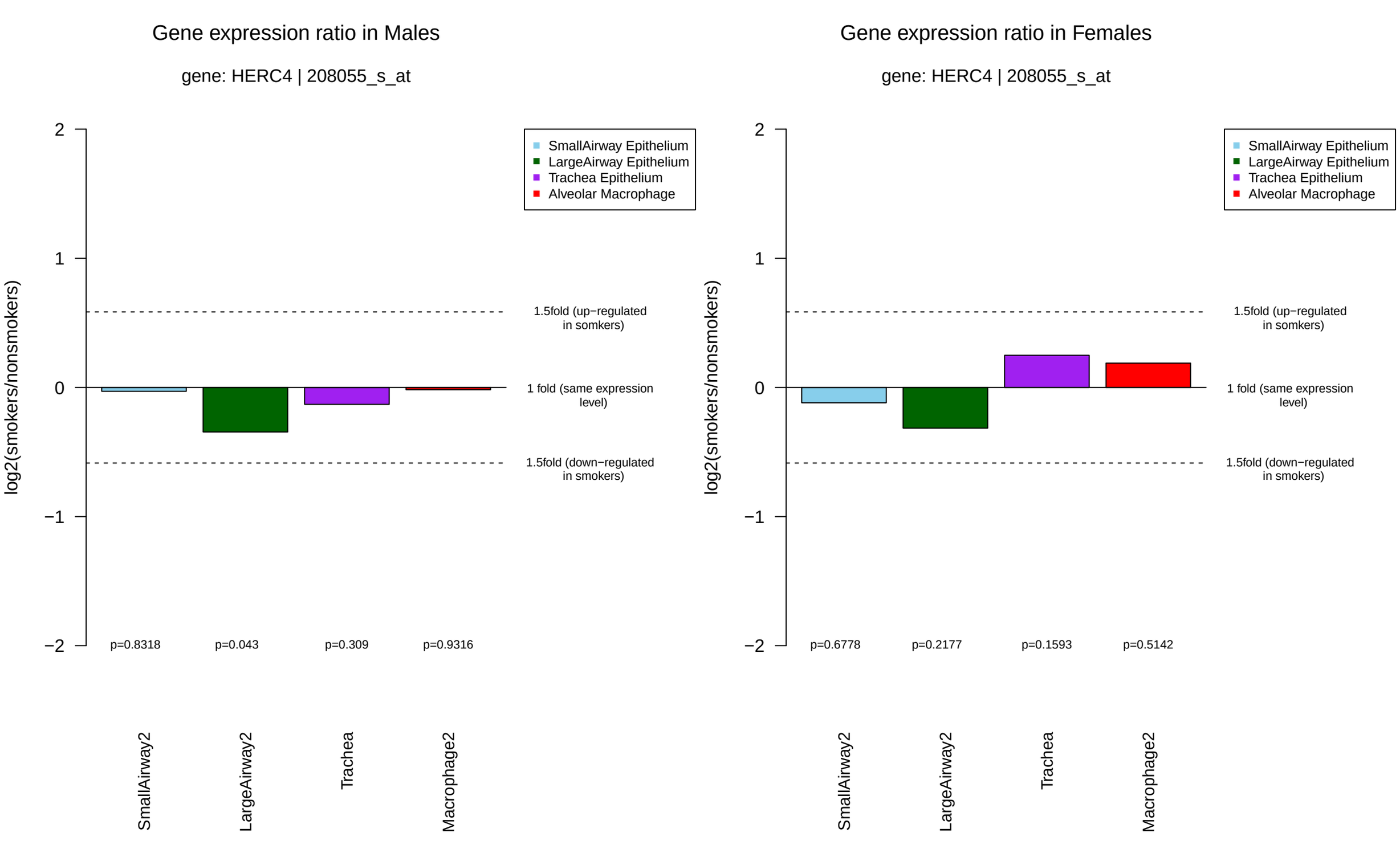
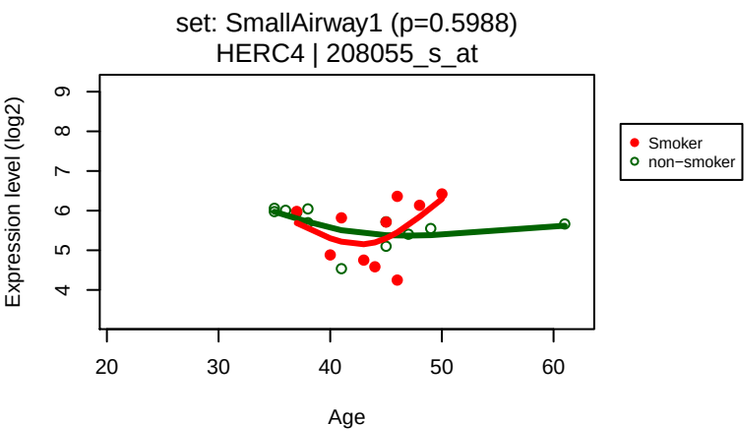
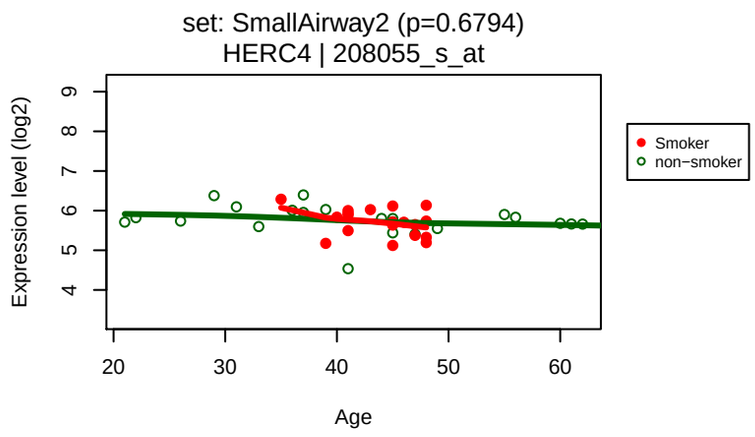
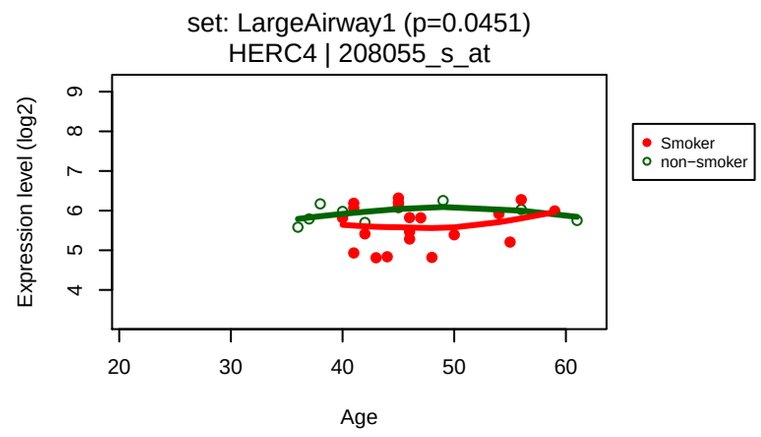

Gene expression of smokers and nonsmokers
gene: HERC4 | 208055_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5988
p=0.6794
p=0.0527
p=0.0451
p=0.0085
p=0.7195
p=0.6306
p=0.8141
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

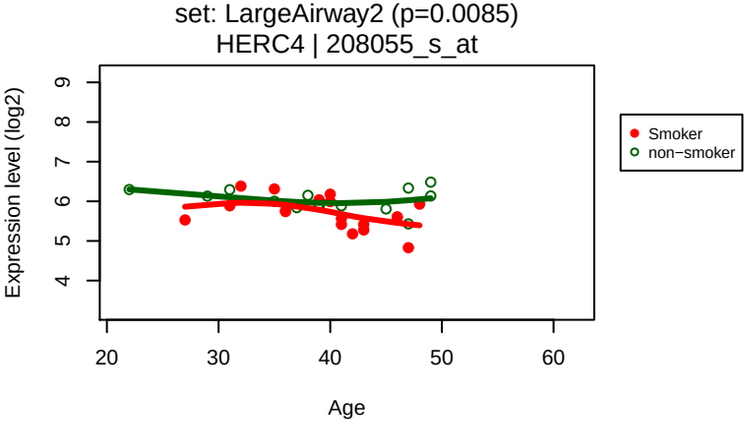

Gene expression ratio (smokers/nonsmokers)
gene: HERC4 | 208055_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5988
p=0.6794
p=0.0527
p=0.0451
p=0.0085
p=0.7195
p=0.6306
p=0.8141
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

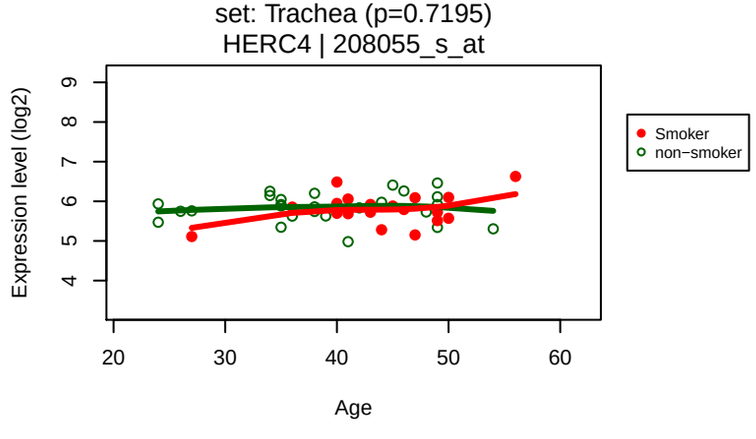

Gene expression ratio in Males
gene: HERC4 | 208055_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8318
p=0.043
p=0.309
p=0.9316
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HERC4 | 208055_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6778
p=0.2177
p=0.1593
p=0.5142
SmallAirway2
LargeAirway2
Trachea
Macrophage2

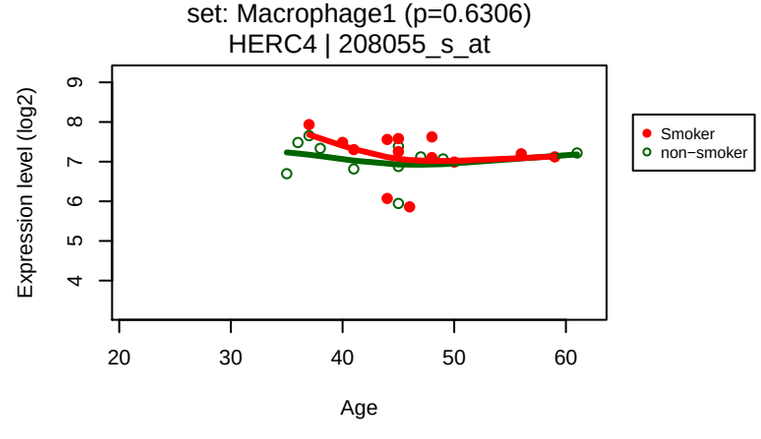

set: SmallAirway1 (p=0.5988)
HERC4 | 208055_s_at
Expression level (log2)
Age
Smoker
non−smoker

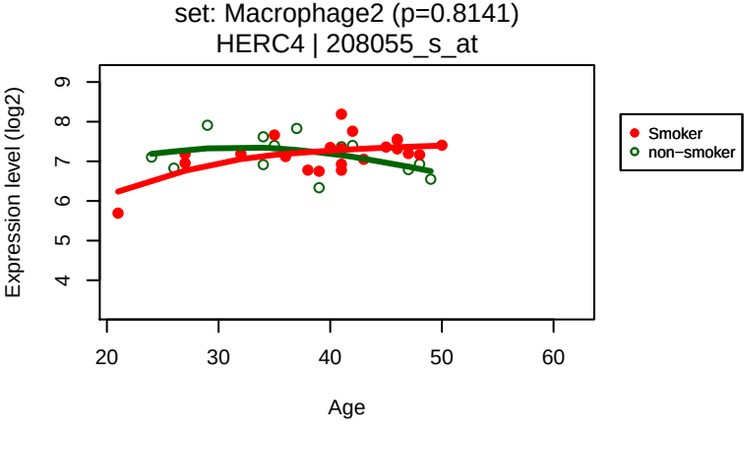

set: SmallAirway2 (p=0.6794)
HERC4 | 208055_s_at
Expression level (log2)
Age
Smoker
non−smoker

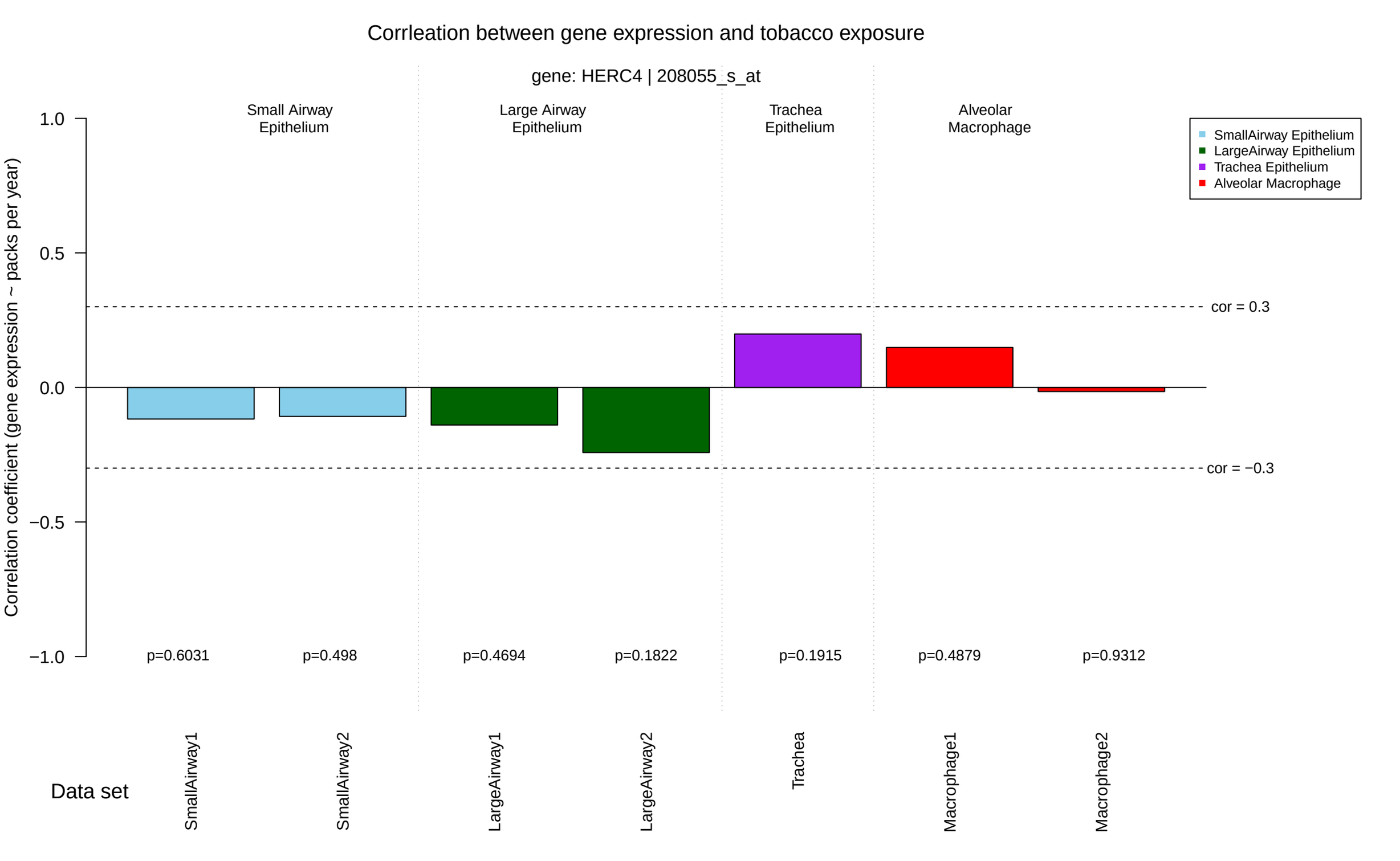

set: LargeAirway1 (p=0.0451)
HERC4 | 208055_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0085)
HERC4 | 208055_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7195)
HERC4 | 208055_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6306)
HERC4 | 208055_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8141)
HERC4 | 208055_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HERC4 | 208055_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6031
p=0.498
p=0.4694
p=0.1822
p=0.1915
p=0.4879
p=0.9312
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2